



SwiftCourier

DBMS Project · Course 25TU03MJM0

Courier Parcel Booking and Tracking Management System

A PHP + MySQL web app: customers book parcels, admins update their status, and anyone can follow a parcel live by its tracking ID or by scanning its QR code.

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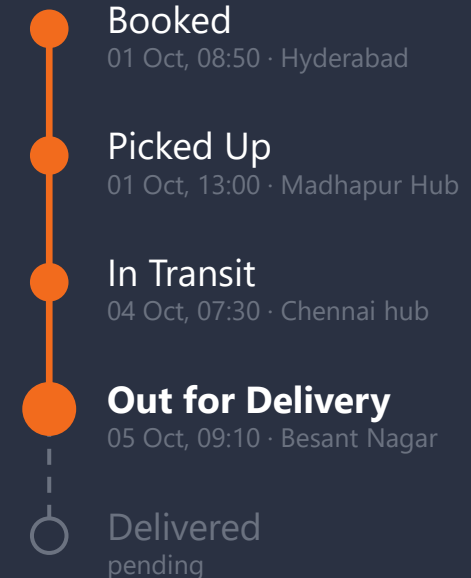
Jogi Dhanush

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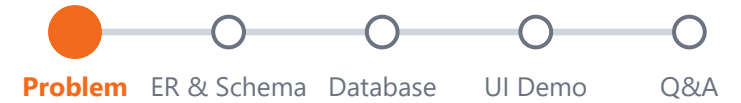
LIVE TRACKING

CR202610010001

Hyderabad → Chennai · Package



Courier counters still run on paper



PROBLEM STATEMENT

Small and mid-sized courier operators still book parcels in paper registers and spreadsheets. No one can see a parcel's full journey, charges are worked out by hand, and a customer has to phone the branch to ask where the parcel is.

OUR ANSWER

One relational MySQL database behind a PHP web app. Customers book online, staff update status, every change is kept as tracking history, and a QR code on each parcel opens its live status.



Paper registers

Bookings live in books and spreadsheets, so the same parcel is recorded twice, differently.



No visibility

A parcel's status lives in phone calls, not in a system a customer can check.



Manual billing

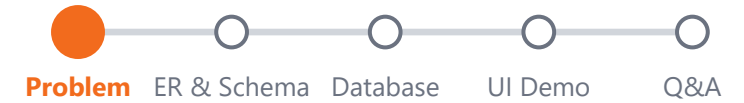
Charges are worked out by hand, with no link to the parcel's weight or type.



No history

When a status is overwritten, the earlier steps of the journey are lost.

Objectives, scope and intended users



Objectives

1. Book a parcel online, with the charge calculated automatically
2. Give every parcel a unique tracking ID and a QR code
3. Store every status change as a new history row
4. Normalise the design to 3NF and enforce it with constraints
5. Show admins live statistics straight from SQL

Scope

IN SCOPE

- Customer accounts and admin roles
- Parcel booking and charges
- Tracking history and public tracking
- QR codes, simulated payments, dashboard

OUT OF SCOPE (FUTURE)

- Real payment gateway
- Live GPS location
- SMS and email alerts

Intended users



Customer

registers, books and tracks parcels



Admin / staff

updates status, location and remarks



Public visitor

tracks by ID or QR scan, no login



Receiver

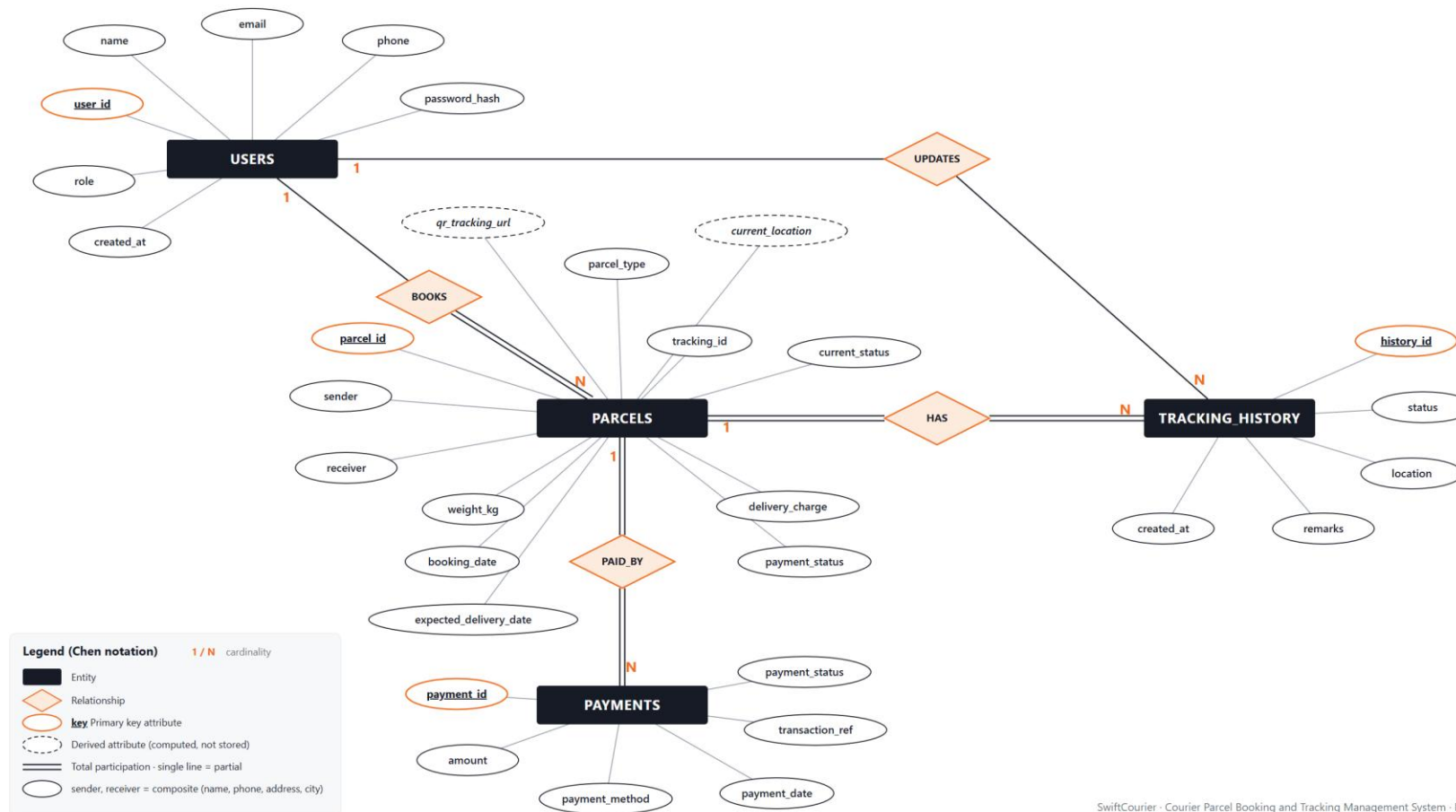
scans the QR to see delivery status



Manager

reads the live dashboard statistics

ER diagram: 4 entities, 4 relationships



4 entities

PARCELS is the hub: users book it, history and payments belong to it

4 relationships

all one-to-many (1 : N)

2 derived attributes

current_location and the QR URL are computed, never stored

2 composite attributes

sender, receiver = name + phone + address + city

SwiftCourier - Courier Parcel Booking and Tracking Management System - ER Diagram

Cardinality and participation rules

Relationship	Entities	Cardinality	Participation
BOOKS	USERS – PARCELS	1 : N	Parcel total
HAS	PARCELS – TRACKING_HISTORY	1 : N	Both total
UPDATES	USERS – TRACKING_HISTORY	1 : N	Partial (FK can be NULL)
PAID_BY	PARCELS – PAYMENTS	1 : N	Both total



Total = NOT NULL foreign key

parcels.customer_id, tracking_history.parcel_id and payments.parcel_id are NOT NULL, so a child row cannot exist without its parent.



HAS is total on both sides

The booking transaction inserts each parcel together with its first 'Booked' history row, so no parcel is ever without history.



Partial = nullable foreign key

tracking_history.updated_by can be NULL (ON DELETE SET NULL), so history survives if an admin account is removed.



Why PAID_BY is 1 : N, not 1 : 1

Every booking creates exactly one payment row today, but the design also allows a retry or a refund row later without changing the schema.

Relational schema: 4 tables and 1 view

users

PK user_id

UQ email

name, phone, password_hash, role (ENUM), created_at

VIEW

v_parcel_overview

parcels JOIN users, plus the latest location from a subquery

parcels

PK parcel_id

UQ tracking_id

FK customer_id → users

sender_name, sender_phone, sender_address, sender_city

receiver_name, receiver_phone, receiver_address, receiver_city

parcel_type, weight_kg, booking_date, expected_delivery_date, delivery_charge, payment_status, current_status, created_at

tracking_history

PK history_id

FK parcel_id → parcels

FK updated_by → users

status, location, remarks, created_at

payments

PK payment_id

FK parcel_id → parcels

UQ transaction_ref

amount, payment_method, payment_status, payment_date



Surrogate keys + business keys: integer PKs keep joins small and stable; tracking_id, email and transaction_ref are the UNIQUE values people actually see. The QR code is never stored: it is generated from tracking_id.

Normalised step by step to 3NF (and BCNF)

STARTING POINT: THE PAPER REGISTER

```
BOOKING( tracking_id, customer_name, customer_email, sender..., receiver..., weight, charge, status_1, location_1, status_2, location_2 ..., pay_mode, txn_ref )
```

1NF

Atomic values, no repeating groups

status_1, location_1, status_2 ... is a repeating group → one row per event in TRACKING_HISTORY.
Every column holds a single value.



2NF

No partial dependencies

Keyed by (tracking_id, event time), the sender, receiver and weight depend on tracking_id alone → they stay in PARCELS; events get their own key, history_id.



3NF

No transitive dependencies

customer_id → name, email, phone ⇒ USERS.
transaction_ref → amount, method ⇒ PAYMENTS.
current_location and the QR URL are derived ⇒ computed, not stored.

Result: 4 tables in 3NF, and BCNF because every determinant is a key. current_status and payment_status are kept as summary copies for fast dashboards, updated in the same transaction so they never disagree.

Database creation with full constraints

courier_management.sql

```
CREATE TABLE parcels (  
  parcel_id      INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  tracking_id    CHAR(14) NOT NULL,  
  customer_id    INT UNSIGNED NOT NULL,  
  ...            -- sender + receiver columns  
  parcel_type    ENUM('Document','Package','Electronics',  
                      'Fragile','Other') NOT NULL,  
  weight_kg      DECIMAL(6,2) NOT NULL,  
  current_status ENUM('Booked', ... , 'Returned')  
                  NOT NULL DEFAULT 'Booked',  
  CONSTRAINT uq_parcels_tracking UNIQUE (tracking_id),  
  CONSTRAINT fk_parcels_customer FOREIGN KEY (customer_id)  
    REFERENCES users (user_id) ON DELETE RESTRICT,  
  CONSTRAINT chk_parcels_weight  
    CHECK (weight_kg > 0 AND weight_kg <= 50)  
);
```

4

**PRIMARY
KEY**

4

**FOREIGN
KEY**

3

CHECK

3

UNIQUE

7

**ENUM
DOMAINS**

1

VIEW

Referential actions

CASCADE: deleting a parcel removes its history and payments.

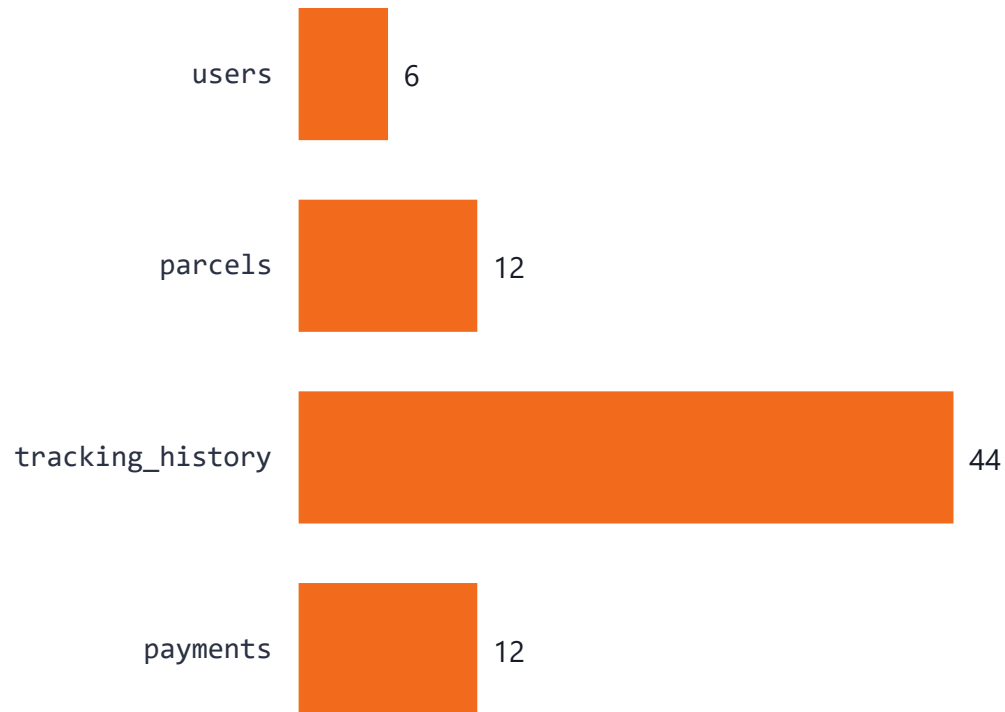
RESTRICT: a customer who still has parcels cannot be deleted.

SET NULL: removing an admin keeps the history rows.

Sample data and integrity checks

Rows inserted per table

Minimum required: 5 per table



RUN LIVE: EACH OF THESE MUST FAIL

ERROR 4025

CHECK

Book a 75 kg parcel (limit is 50 kg)

ERROR 1452

FOREIGN KEY

Book a parcel for customer_id 99, who does not exist

ERROR 1062

UNIQUE

Register customer@courier.com a second time

ERROR 1451

RESTRICT

Delete a customer who still has parcels

ERROR 1265

ENUM (strict mode)

Set a parcel's status to 'Lost'



XAMPP's MariaDB reports a failed CHECK as 4025; MySQL 8 reports the same rule as 3819.

Tracking a parcel is a 3-table join

Q9 · TRACK A PARCEL

```
SELECT th.created_at, th.status, th.location,  
       th.remarks, u.name AS updated_by  
FROM tracking_history th  
JOIN parcels p      ON p.parcel_id = th.parcel_id  
LEFT JOIN users u   ON u.user_id  = th.updated_by  
WHERE p.tracking_id = 'CR202610010001'  
ORDER BY th.created_at;
```

Also in the app

The admin dashboard joins parcels with users for Recent bookings. The Customers page uses LEFT JOIN ... GROUP BY, so customers with zero bookings still appear.

RESULT · 5 ROWS

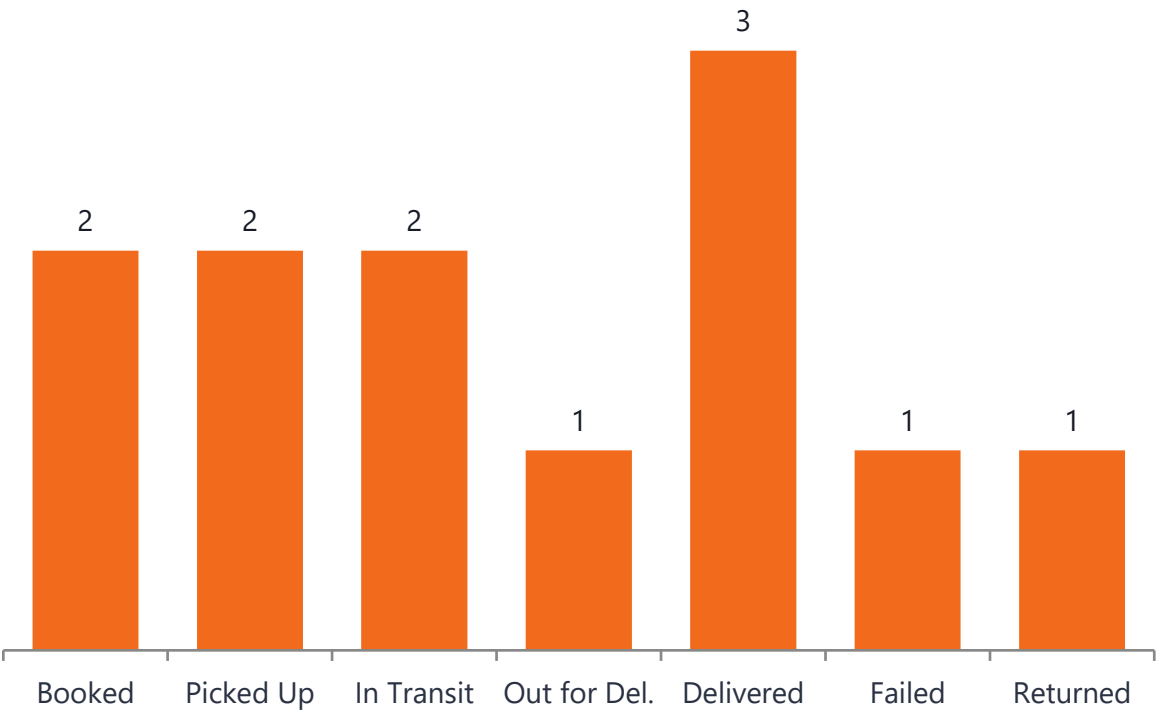
created_at	status	location	updated_by
01 Oct 08:50	Booked	Hyderabad	Rahul Sharma
01 Oct 13:00	Picked Up	Madhapur Hub	Admin User
02 Oct 06:00	In Transit	Hyderabad Sorting	Admin User
04 Oct 07:30	In Transit	Chennai Sorting	Ops Staff
05 Oct 09:10	Out for Delivery	Basant Nagar	Ops Staff

Hyderabad → Chennai: the same rows the tracking page and the QR scan draw as a timeline. Locations shortened to fit.

Aggregates turn bookings into reports

Parcels by current status

COUNT(*) with GROUP BY current_status



BOOKINGS PER CUSTOMER · LEFT JOIN + GROUP BY

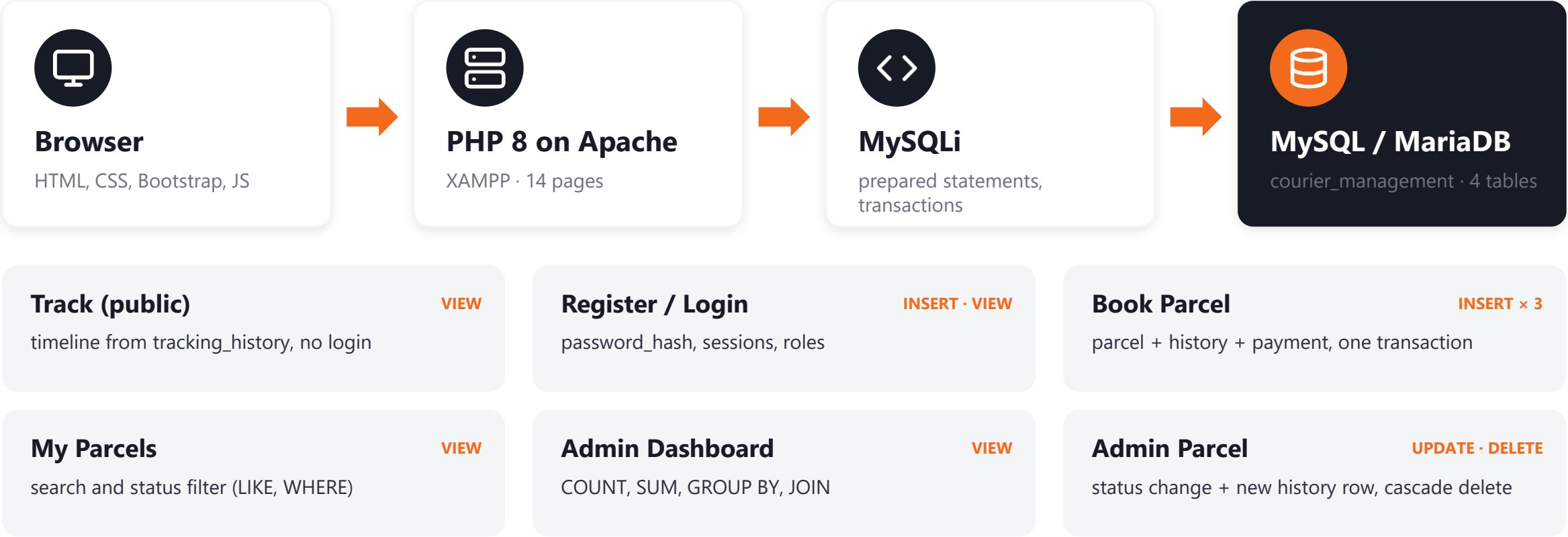
Rahul Sharma	₹500	5
Priya Nair	₹200	3
Arjun Mehta	₹300	2
Sneha Reddy	₹204	2

THE CHART'S QUERY

```
SELECT current_status, COUNT(*)  
FROM parcels  
GROUP BY current_status;
```

Revenue SUM(amount) WHERE payment_status = 'Paid': ₹780.
HAVING COUNT(*) > 1 lists customers with repeat bookings.

A web UI wired straight to MySQL



 **Every number is live SQL:** the admin dashboard has a panel that shows the exact queries behind its cards.

Insert: one booking writes 3 tables

1 Customer fills the booking form

SwiftCourier

DashboardBook ParcelMy ParcelsTrackRahul

Book a Parcel

Fill in the details below. You'll get a tracking ID and QR code instantly.

Sender details

Sender name *
Rahul Sharma

Sender phone *
9876543210

Pickup address *
Flat 302, Green Residency, Madhapur

City *
Hyderabad

Receiver details

Receiver name *
Meena Iyer

Receiver phone *
9830012345

Delivery address *
14, Park Street

City *
Kolkata

Parcel details

Parcel type *
Fragile (+₹40)

Weight (kg) *
2.5

Expected delivery *
10-10-2026

Charges

Base charge ₹50.00
2.5 kg x ₹20 ₹50.00
Type surcharge ₹40.00
Total ₹140.00

Charge = ₹50 base + ₹20/kg + type surcharge. The server recalculates it when you book.

Payment method *
UPI

① Simulated payment for this academic project: online methods are marked Paid; Cash on Pickup stays Pending until delivery.

Confirm Booking

2 Booked: tracking ID, charge and QR

SwiftCourier

DashboardBook ParcelMy ParcelsTrackRahul

Parcel booked successfully! Your tracking ID is CR202610060001.

Booking confirmed!
Tracking ID **CR202610060001**
Status: Booked · Delivery charge: ₹140.00 · Payment: Paid

My Parcels

Parcel CR202610060001

BookedPublic tracking page

Shipment details

Booked 06 Oct 2026, 08:24 AM

From Hyderabad To Kolkata

Sender
Rahul Sharma
9876543210
Flat 302, Green Residency, Madhapur, Hyderabad

Receiver
Meena Iyer
9830012345
14, Park Street, Kolkata

PARCEL TYPE	WEIGHT	EXPECTED DELIVERY	DELIVERY CHARGE
Fragile	2.5 kg	10 Oct 2026	₹140.00

Payment

Paid

REFERENCE	METHOD	AMOUNT	STATUS	DATE
TXN202610060824410013	UPI	₹140.00	Paid	06 Oct 2026, 08:24 AM

Parcel QR code

CR202610060001
Scan to track: http://192.168.0.104/courier/track.php?tracking_id=CR202610060001
ViewDownloadPrint

parcels

12 → 13
INSERT

tracking_history

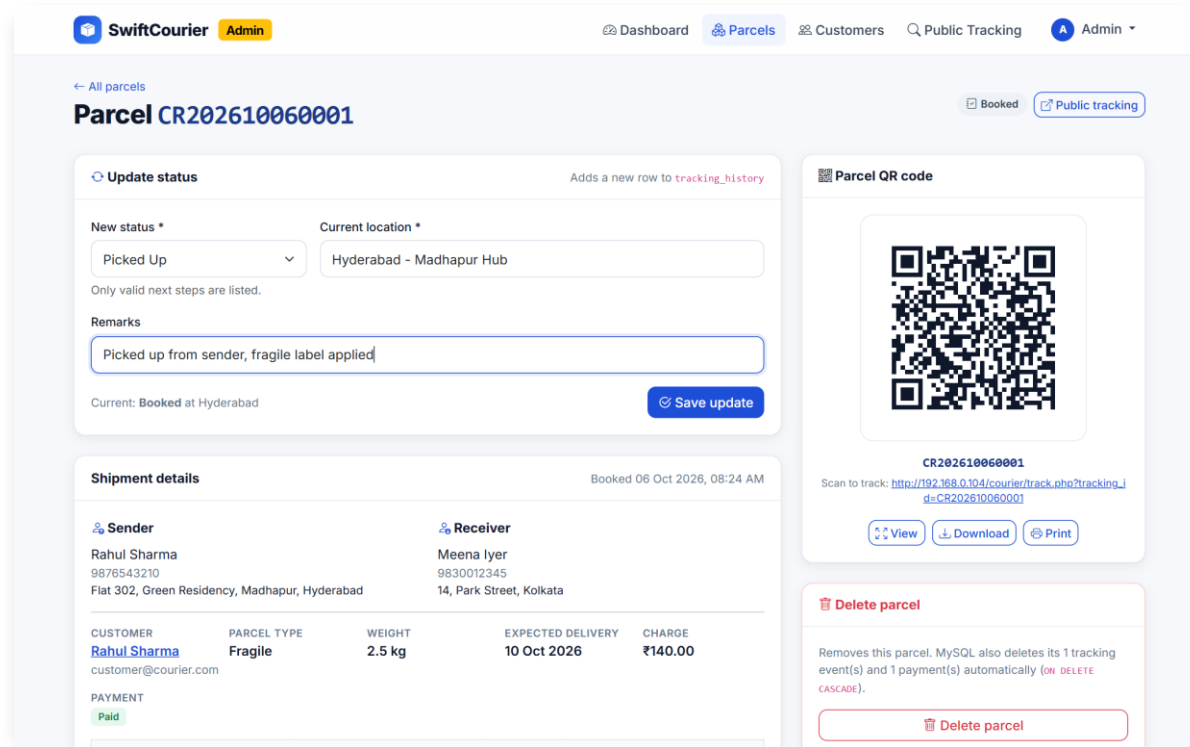
44 → 45
'Booked' row, same transaction

payments

12 → 13
INSERT, same transaction

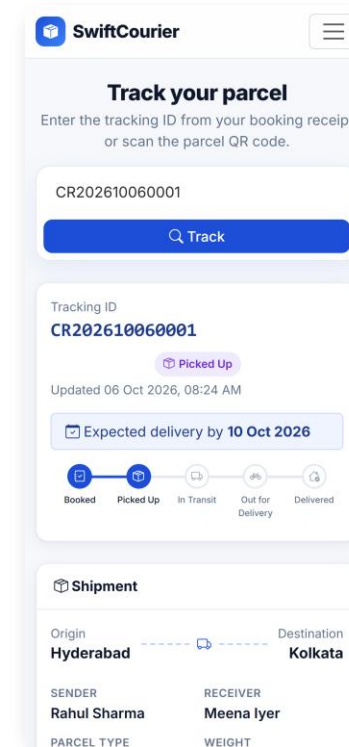
Update: every status change is a new row

1 Admin sets Picked Up at Madhapur Hub



The screenshot shows the SwiftCourier Admin interface. At the top, there's a navigation bar with 'Dashboard', 'Parcels', 'Customers', 'Public Tracking', and 'Admin'. The main content area is titled 'Parcel CR202610060001'. It has a 'Public tracking' link. Below this, there's an 'Update status' section with a dropdown for 'New status' (set to 'Picked Up') and a text field for 'Current location' (set to 'Hyderabad - Madhapur Hub'). There's a 'Remarks' field with the text 'Picked up from sender, fragile label applied'. A 'Save update' button is at the bottom right. Below the update section, there's a 'Shipment details' section showing 'Sender' (Rahul Sharma) and 'Receiver' (Meena Iyer) information. At the bottom, there's a table with columns: CUSTOMER, PARCEL TYPE, WEIGHT, EXPECTED DELIVERY, and CHARGE. The table shows 'Rahul Sharma', 'Fragile', '2.5 kg', '10 Oct 2026', and '₹140.00'. There's also a 'Delete parcel' button at the bottom right.

2 Phone scans the QR



The screenshot shows the SwiftCourier mobile app interface. It has a 'Track your parcel' section with a text field for 'Enter the tracking ID from your booking receipt or scan the parcel QR code.' Below this, there's a 'Track' button. The tracking ID is 'CR202610060001'. There's a 'Picked Up' status indicator. Below this, there's a 'Tracking ID' section with the ID 'CR202610060001'. There's a 'Expected delivery by 10 Oct 2026' section. At the bottom, there's a 'Shipment' section showing 'Origin' (Hyderabad) and 'Destination' (Kolkata). There's a 'SENDER' (Rahul Sharma) and 'RECEIVER' (Meena Iyer) section. There's a 'PARCEL TYPE' and 'WEIGHT' section.

ONE TRANSACTION

```
START TRANSACTION;
UPDATE parcels
  SET current_status
    = 'Picked Up';
INSERT INTO
  tracking_history
VALUES (...);
COMMIT;
```

parcels

Booked → Picked Up
UPDATE current_status

tracking_history

45 → 46
INSERT, old rows kept

QR scan

live status
opens track.php on the phone

Delete: cascades or refuses, by design

DELETE parcel CR202610060001

SwiftCourier Admin

Dashboard Parcels Customers Public Tracking Admin

Deleted parcel CR202610060001. MySQL also removed its 2 tracking event(s) and 1 payment(s) automatically (ON DELETE CASCADE).

Parcels

12 parcel(s) found.

Tracking ID, customer, sender, receiver or phone All statuses Filter

TRACKING ID	CUSTOMER	RECEIVER	ROUTE	TYPE	BOOKED	CHARGE	STATUS	ACTION
CR202610040003	Rahul Sharma 9876543210	Suresh Babu 9866012345	Hyderabad → Warangal	Package 2 kg	04 Oct 2026	₹90.00 Pending	Failed	Manage
CR202610040002	Priya Nair 9876500011	Harish Kumar 9884012345	Chennai → Coimbatore	Document 0.2 kg	04 Oct 2026	₹54.00 Paid	Booked	Manage
CR202610040001	Sneha Reddy 9876500033	Aditi Singh 9829012345	Bengaluru → Jaipur	Package 1.2 kg	04 Oct 2026	₹74.00 Pending	Booked	Manage
CR202610030002	Rahul Sharma 9876543210	Deepak Varma 9849012345	Hyderabad → Visakhapatnam	Fragile 1 kg	03 Oct 2026	₹110.00 Paid	Picked Up	Manage
CR202610030001	Arjun Mehta 9876500022	Neha Joshi 9820012345	Pune → Mumbai	Other 5 kg	03 Oct 2026	₹150.00 Pending	Picked Up	Manage
CR202610020001	Priya Nair 9876500011	Sanjay Gupta 9948012345	Kochi → Hyderabad	Document 0.3 kg	02 Oct 2026	₹56.00 Paid	In Transit	Manage
CR202610010002	Sneha Reddy 9876500033	Rohit Malhotra 9810098100	Bengaluru → New Delhi	Electronics 2.5 kg	01 Oct 2026	₹130.00 Paid	In Transit	Manage



CASCADE: its 2 tracking events and 1 payment go too. parcels 13 → 12, tracking_history 46 → 44

DELETE customer Rahul Sharma

SwiftCourier Admin

Dashboard Parcels Customers Public Tracking Admin

MySQL refused to delete Rahul Sharma: they still have 5 parcel(s). The foreign key parcels.customer_id is ON DELETE RESTRICT (error 1451), so booking records cannot be orphaned.

Customers

4 registered customer(s) with booking counts.

Search name, email or phone

CUSTOMER	PHONE	JOINED	BOOKINGS	ACTIVE	DELIVERED	TOTAL CHARGES	LAST BOOKING	ACTIONS
Rahul Sharma customer@courier.com	9876543210	05 Sep 2026	5	2	1	₹500.00	04 Oct 2026	View
Priya Nair priya@example.com	9876500011	08 Sep 2026	3	2	1	₹200.00	04 Oct 2026	View
Arjun Mehta arjun@example.com	9876500022	12 Sep 2026	2	1	1	₹300.00	03 Oct 2026	View
Sneha Reddy sneha@example.com	9876500033	15 Sep 2026	2	2	0	₹204.00	04 Oct 2026	View

SwiftCourier — Courier Parcel Booking & Tracking Management System DBMS Project · PHP + MySQL



RESTRICT: MySQL error 1451, because this customer still has 5 parcels

Delivered.

Thank you. Questions?

- ✓ 4-table schema in 3NF and BCNF, with 4 FKs, 3 CHECKs and 7 ENUMs
- ✓ Transactions keep parcels, history and payments in step
- ✓ 13 demo queries: JOIN, LEFT JOIN, GROUP BY, HAVING, a view
- ✓ Working PHP app: insert, update, delete, view, and QR tracking on a phone

GitHub: github.com/akkalasrestadeepgoud333-code/DBMS-Course-Project

NEXT STOP: FUTURE SCOPE

-  **SMS and email alerts**
at every status change
-  **Real payment gateway**
UPI and cards instead of simulation
-  **Live GPS tracking**
the delivery agent's location on a map
-  **Proof of delivery**
photo and OTP in an agent app